Password-Protected BLUETOOTH REMOTE CONTROL

TEJ VIJAYKUMAR PATEL

This password-protected wireless remote control operating at 2.4GHz frequency can be used to control robots, home appliances, machines, etc. Here we have used the secure remote to control a war robot that has three motors, one of which is used to lift objects or weapons such as a cutter (or sword). The article describes configuration of the remote control system and doesn't cover construction and working of the robot in detail due to paucity of space.

Circuit and working

The project consists of two pairs of Arduino and Bluetooth HC-05 modules, one of which is used in the transmitter unit and the other in the receiver unit. The Bluetooth module on the transmitter side is configured as master, while the module on the receiver side is configured as slave. The author's prototypes for the transmitter unit and the receiver unit are shown in Figs 1 and 2, respectively.

Transmitter unit. Circuit diagram of the transmitter unit is shown in Fig. 3. The prototype uses six push-buttons/switches (S1 through S6), which are connected to the Arduino board (Board1) using jumper wires. All these switches are used to control the motors for different movements of the robot. The HC-05 Bluetooth module (BT1) is also connected to Board1 as shown in Fig. 3.

Receiver unit. Circuit diagram of the receiver unit is shown in Fig. 4. It consists of Arduino board (Board2), HC-05 Bluetooth module (BT2), six 12V, 1CO relays (RL1 through RL6), three DC motors (M1, M2 and M3), six relay driver transistors 2N2222 (T1 through T6) and a few other components. The author's prototype uses a 4-channel relay card for the robot's right and left wheels and a 2-channel relay card to drive the motor for lifting the object. BT2 is connected to Board2 using jumper wires. All the relay cards are connected to Board2.

The relay card should be made

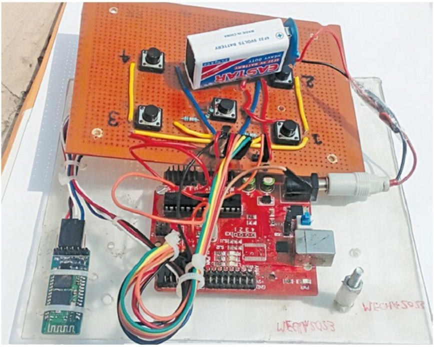

Fig. 1: Author's prototype of transmitter unit

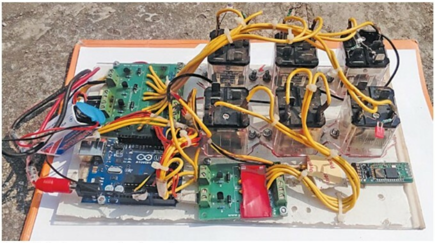

Fig. 2: Author's prototype of receiver unit